

Identifying Temperature Condition and Risk-Taking Scores with Neural and Physiological Features

Blake Turman

ID: 11652799

Abstract— Non-invasive techniques to measure brain activity, such as electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS), can add a wall of depth to environmental and behavioral neuroscience research. However, analyzing complex multi-feature EEG and fNIRS data can introduce particular difficulty. This work proposes to analyze combined EEG and fNIRS data using artificial neural networks to determine the neural and neurophysiological features that link environmental temperature to risk-taking behavior. The method involves two shallow Multi-Layer Perceptron (MLP) models trained on EEG and fNIRS data. One model predicts hot vs neutral condition. The other model predicts risk-taking behavior. The results are compared to find the link between temperature condition and risk-taking behavior based on the input features.

I. INTRODUCTION

Electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) are non-invasive technologies used for studying brain activity. These technologies can serve to identify neural or neurophysiological features associated with certain behaviors. It is understood that environmental temperature conditions can affect risk-taking behavior [1]. EEG and fNIRS have shown to be able to identify the neural and neurophysiological markers involved in temperature response [2][3] and risk-taking behavior [4][5][6].

Artificial neural networks are learning algorithms which can self-optimize function parameters to represent multi-feature data nonlinearly with the goal of making accurate predictions. This makes them very useful for extracting meaning from complex data, and a great solution for analyzing the complex features of EEG and fNIRS data. Past research has found predictive success using shallow feedforward networks for EEG and fNIRS data [7][8]. Other research has found combining EEG and fNIRS data leads to improved prediction outcomes [9][10][11]. The focus of this work is to analyze EEG and fNIRS data, utilizing shallow feedforward networks, to identify the neural and physiological signatures linking environmental temperature and risk-taking behavior.

II. METHODS

A. Data Acquisition

The data comes from the WSU SCENe lab Temperature and Cognition study. Participants are placed inside of a sauna and set to perform a series of tasks. Each participant is scheduled on two different days to be placed in both the hot (heated sauna) condition and neutral (sauna off) condition. This analysis focuses on the Balloon Analogue Risk Task (BART)—a computerized task designed to assess risk-taking behavior [12]. Each participant must complete BART while inside of the sauna and task metrics are automatically recorded.

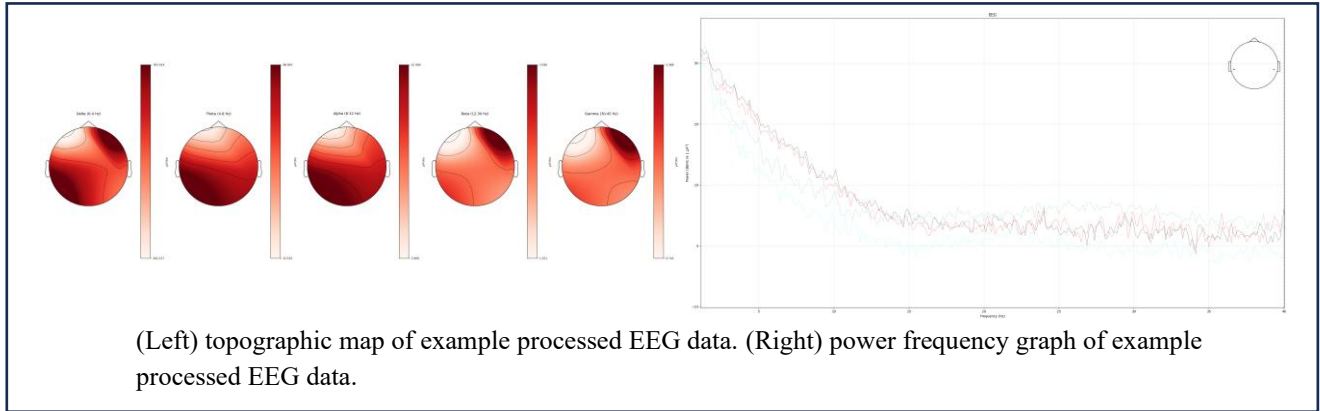
Input data is taken from the MUSE S Athena EEG+fNIRS headband worn by each participant. The headband has 4 EEG electrodes (TP9, Fp1, Fp2, TP10) and 4 fNIRS optodes (left 3 cm, left 1 cm, right 1

cm, right 3 cm). Each optode records at two wavelengths (730 nm, 850 nm). Raw EEG and fNIRS timeseries data are downloaded from the device. All data is epoched per task.

B. Preprocessing

Raw data is processed using the MNE-Python and MNE-NIRS libraries for processing neurophysiological data. EEG data is filtered from 1-40 Hz to isolate neuronal electrical activity [13]. fNIRS data is normalized by the log-ratio optical density conversion, cleaned with temporal derivative distribution repair, and enhanced with short channel regression [14][15][16]. Data is then converted to hemoglobin concentration using Beer-Lambert's law, and finally filtered from 0.01-0.1 Hz to isolate cerebral blood fluctuations [17]. Within-channel normalization leads to improved outcomes for supervised learning tasks [18]; thus, each electrode channel is additionally z-score normalized. EEG data is separated by frequency band and averaged, resulting in scalars of frequency bands per each electrode. fNIRS data is also averaged, resulting in scalars of hemoglobin oxygenation state per each optode.

To simplify the gradients of the model, all input data is z-score normalized prior to inputting. Additionally, temperature condition is encoded via a single binary indicator, and BART scores are encoded via average relative risk, taken as $\frac{1}{n} \sum \frac{\text{number risky decisions}}{\text{possible risky decisions}}$.



C. Neural Network Models

A total of 454 samples were obtained, 30 of these samples being BART task samples. Due to small n sample size, shallow feedforward networks are best suited [19]. Two Multi-Layer Perceptron (MLP) models are created for comparison between temperature condition and BART performance. Both are implemented using PyTorch. The temperature condition model has 1 input layer, 2 fully-connected hidden layers with Rectified Linear Unit (ReLU) activation, and 1 output layer with sigmoid activation. The BART performance model has 1 input layer, 1 hidden layer, and 1 output layer with sigmoid activation. Due to concerns over sample efficiency, the BART performance model uses sigmoid activation instead of ReLU activation for its hidden layer [20][21].

Following standardization guidelines for applying machine learning to neurophysiological data [8], each MLP utilizes nested cross validation. The validation loop is implemented using Skorch and Scikit-learn. Hyperparameters are tuned using grid search. The temperature condition model grid search is scored with classification accuracy. The BART performance model grid search is scored with R^2 value.

Searchable hyperparameters are the same for both models and include learning rate, hidden size, max epochs, batch size, and weight decay.

To ensure no participant overlap in training and testing sets, the both models employ Group K-Fold validation, grouped by participant. The temperature condition validation loop makes 5 splits for the inner fold and 10 splits for the outer fold. The BART performance model makes 3 splits for the inner fold and 5 splits for the outer fold.

For such small sample sizes, it is important to consider dimensionality. Thus, input data is reduced using Select-K-Best feature selection within the BART performance model validation loop. Mutual Information (MI) regression is used as the scoring function for feature selection. The same selected features from the BART performance model are used for the temperature condition model to allow for cross-model analysis.

Both models are trained using Stochastic Gradient Descent (SGD) with mini-batching. To improve performance considering small sample sizes, Nesterov Accelerated Gradient (NAG) is used with momentum of 0.9 [22]. The temperature condition model employs Binary Cross-Entropy (BCE) loss. The BART performance model employs Mean Squared Error (MSE) loss. Both models utilize L2 (Ridge) Regularization. Following the validation loop, each model is trained on all available data using the best features and hyperparameters selected from the validation loop.

D. Baselines

To verify the relevance of the method, linear baseline comparisons for each model are made. The temperature condition model uses the default logistic regression from Scikit-learn as a baseline, while the BART performance model uses a fractional logistic regression made from Scikit-learn linear regression, as well as logit and expit functions as a baseline. Each baseline is inserted into a validation loop in likewise fashion to the MLP models. The baseline validation loops are kept as close to the same as possible to their nonlinear counterparts, however, the linear models do not possess the equivalent hyperparameters. Because of this, grid search searchable hyperparameters only include C inverse regularization strength for the temperature condition baseline, and fit y -intercept for the BART performance baseline. Additionally, NAG is not used. Baseline performances are compared to the MLP models to determine the value of a nonlinear approach to analyzing the EEG and fNIRS data.

E. Analysis

Since both models utilize the BART task samples as training input, each can be compared via these samples. For analysis, the temperature condition model is trained on all 454 samples and the BART performance model is trained on all 30 BART samples. Then to cross-analyze, each model makes predictions on the 30 BART samples.

Two strategies are applied for determining the relationship between temperature condition and BART performance based on the neural and physiological data. The first strategy involves simply taking the correlation between the output of each model for the same block. This directly determines the relatedness of temperature condition and BART performance based on the neural and physiological input. Output correlation is calculated using Pearson product-moment correlation from the NumPy library.

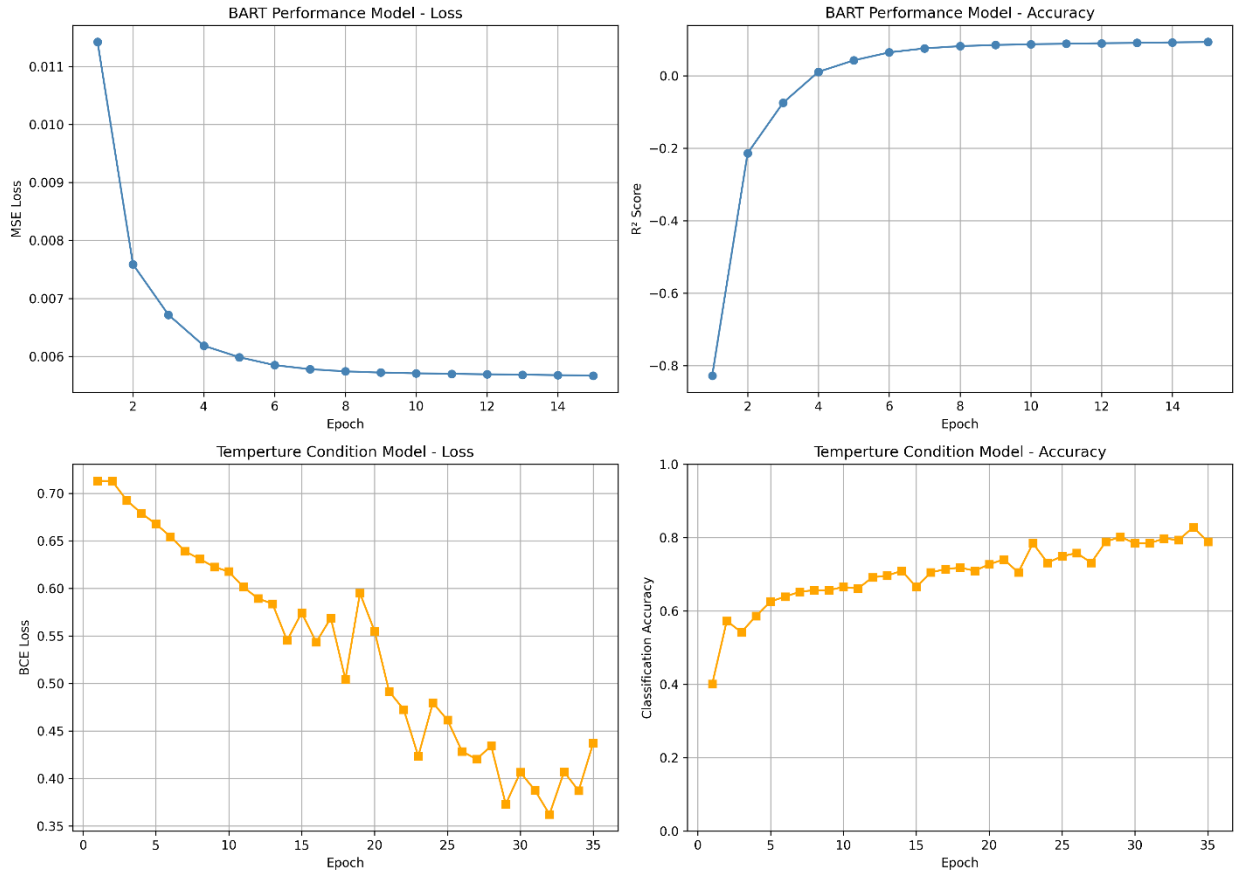
The second strategy involves analyzing and comparing gradient-based saliencies for each feature of the two models. This allows for the determination of feature importance as well as whether features predict both temperature condition and BART performance. Saliencies of the models are calculated for

each block and then absolute values are taken and averaged to obtain a single scalar average magnitude saliency for each feature. Gradients are calculated using the PyTorch backward function.

III. RESULTS

A. Model Comparison

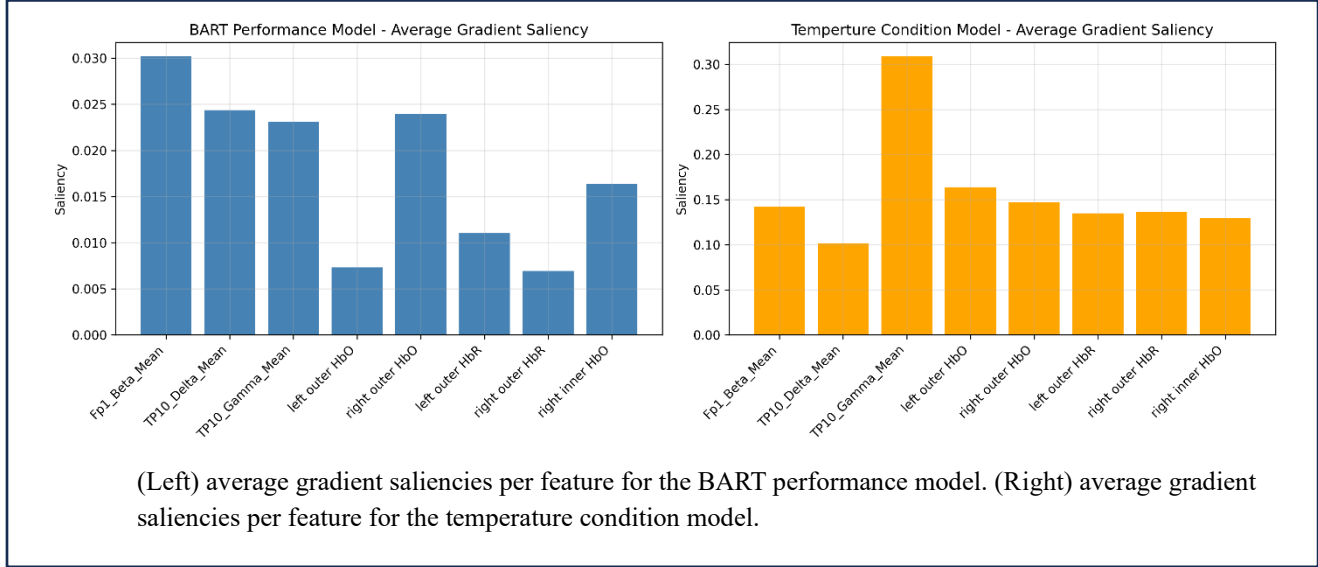
Temperature condition model final best training classification accuracy was .87. BART performance model final best training R^2 score was .18. Output correlation between the two models was .30.



(Top-Left) MSE loss graphed over training epochs of BART performance model. (Top-Right) R^2 accuracy score graphed over training epochs of BART performance model. (Bottom-Left) BCE loss graphed over training epochs of temperature condition model. (Bottom-Right) classification accuracy score graphed over training epochs of temperature condition model.

The BART performance model gradients emphasized the Fp1 Beta EEG channel the most at .030 saliency and deemphasized the left 3 cm (outer) HbO, left 3 cm HbR, and right 3 cm HbR fNIRS channels at .007, .011, and .007 saliencies respectively. TP10 Delta, TP10 Gamma, right 3 cm HbO, and right 1 cm (inner) HbO channels were all relatively important to the model with .024, .023, .024, and .016 saliencies respectively.

The temperature condition model emphasized the TP10 Gamma channel by far the most with an average saliency of .31. All others were given similar importance with Fp1 Beta, left 3 cm HbO, right 3 cm HbO, left 3 cm HbR, right 3 cm HbR, and right 1 cm HbO, having corresponding .14, .16, .15, .13, .14, and .13 saliencies, and the lowest being TP10 Delta with a saliency of .10.



B. Validation Results

The BART performance validation loop resulted in a mean accuracy (R^2 score) of -1.32, a median accuracy of -1.58 and standard deviation accuracy of 1.00. Best accuracy was .24. Selected best features included 3 EEG channels—Fp1 Beta, TP10 Delta, and TP10 Gamma—and 5 fNIRS channels—left 3 cm HbO, right 3 cm HbO, left 3 cm HbR, right 3 cm HbR, and right 1 cm HbO. Best hyperparameters included a batch size of 4, a learning rate of 1, a number epochs of 15, a hidden layer size of 4, and a weight decay of 0.0001.

The temperature condition validation loop showed a mean accuracy (classification) of .54, a median accuracy of .59, and a standard deviation accuracy of .20. Best accuracy was .92. Best hyperparameters included a batch size of 16, a learning rate of 0.1, a number epochs of 35, a hidden layer size of 16, and a weight decay of 0.001.

C. Baseline Results

The BART performance baseline resulted in a mean accuracy of -1.24, a median accuracy of -.89, and a standard deviation accuracy of 1.45. Best accuracy was .19. Selected best features were exactly the same as what the nonlinear BART performance model had selected. Best hyperparameter was fit intercept as true.

The temperature condition baseline led to a mean accuracy of .59, a median accuracy of .66, and a standard deviation accuracy of .14. Best accuracy was .73. Best hyperparameter was inverse regularization strength C of 1.0.

IV. DISCUSSION

Results showed a fairly high positive correlation between the outputs of the models of .30. This means that when the BART performance model saw neurophysiological data that appeared “risky”, the temperature condition model was more likely to also think that that same neurophysiological data appeared “hot”. Because the correlation between risky-taking behavior and temperature was not computed directly but instead computed as the correlation between outputs of the MLP models given the neurophysiological data, this result indicates specifically that there may be certain neurophysiological features associated with increased risk-taking behavior are also associated with heat. Moreover, looking at the gradient saliencies may give a hint as to which neurophysiological features those are. Notable is the TP10 Gamma EEG channel which reigned supreme over all other features in the temperature condition model. Though TP10 Gamma was only the fourth most important feature to the BART performance model, its saliency was still relatively amongst the largest. Additionally, the right 3 cm HbO fNIRS channel, though was not ranked highest for either model, was ranked relatively high for both models. Interestingly, the models seemed to disagree about some of the features, and were given close to opposite levels of importance by each model. For example, the second highest saliency for the temperature condition model (left 3 cm HbO) had the lowest saliency for the BART performance model. Likewise, the second highest saliency for the BART performance model (TP10 Delta) had the lowest saliency for the temperature condition model.

The temperature condition model behaved much as expected, improving its accuracy iteratively as it trained to a point of being able to correctly classify condition more than 80% of the time. The validation loop further indicates an ability for the to generalize to unseen data. The temperature condition model’s best accuracy outperformed the baseline. However, the grid search for this model was more expansive than for the baseline, meaning it had more attempts to arrive a lucky set of parameters. On a more restricted grid search, the linear and nonlinear models perform similarly on average. Thus, the nonlinear approach might not actually be necessary to represent the data well.

The BART performance model exhibited some less favorable behavior. During training, convergence happened very quickly (within a few epochs) to a value slightly above zero. With a bad selection of hyperparameters, the model most often scored close to zero or even negatively. A negative R^2 value means that the model represents the data worse than a horizontal line. Selecting the right hyperparameters was very important to get the model to consistently score above zero, and even then, the best validation score it could muster was only .24. This is only a small improvement over baseline which suggests the model is not learning anything substantially more complex than a linear relationship. There are a few things to say about this. For one, R^2 is harsh scoring mechanism for a neural network, and was selected to maintain a more unbiased representation of model performance. Thus, the low scores may actually be pretty decent given the complexity of the model’s task. There is of course the possibility that the data is just so noisy (i.e. the input data does not significantly correlate with BART scores), that such a simple model could not ever represent it well. EEG and fNIRS timeseries data across the entire task were squashed down to single mean scalars which could have reduced important time-sensitive information to the point of futility. Another possibility is that the sample size was just too small to learn something more meaningful. A sample size of 30 is generally considered too small for even a small neural network to be a worthwhile model of that data. Potentially if the amount of training data was dramatically increased, the model could catch onto a more complex relationship.

A major limitation of this analysis is its reproducibility. This analysis was done for a single run of the validation and training loops. However, results sometimes differed wildly across runs. This was particularly true for the gradient saliencies, as the models emphasized seemingly random features across runs. Significance of the results was not analyzed in this work, and thus their meaningfulness remains somewhat murky.

V. FUTURE WORK

The most obvious direction to take into the future is to continue collecting data. Potentially, stronger results could be encountered if the models are given a large enough dataset to train on. Additionally, it is worth understanding that many of the decisions made for this analysis were based on trying to reduce potential for overfitting on a small dataset. However, as more data becomes available, there is no reason a larger or more complex model with greater capacity couldn't be implemented to replace the current MLP models. More data could also allow for a greater breadth of analysis including an expanded set of features, as there would be less fear of overparameterization. Work into determining the statistical significance of the results could help elucidate which of the results are actually meaningful. Again, a larger sample size could serve an improvement here, as it would help improve generalizability of the models, such that the results may become more reproducible.

Future investigation into the selected features could be a powerful possibility. As the model selects features that are most closely correlated with outputs, it would be interesting to understand why those might have been selected. Similarly, the calculated gradient saliencies for the models indicate which features the model utilizes the most for determining its outputs. Thus, a dive into literature about certain brain regions or brain wave frequencies could provide insightful explanations regarding why the model emphasizes and selects specific features over others.

As it stands, Select-K-Best feature selection is utilized only within the BART performance model. The purpose of doing feature selection in this directive is to answer the question of how risk-taking associated features are also associated with temperature condition. However, there is no reason the reverse analysis couldn't also be performed by simply having the temperature condition model perform the feature selection. This would allow analysis on another set of features, closer associated with temperature.

Lastly, other types of analyses could be performed on this EEG and fNIRS data. For example, the original format of the data is timeseries. Thus, analyses could be performed taking into account this temporal resolution, such as with a convolutional neural network (CNN) or a recurrent neural network (RNN), which are effective for handling tensorized or sequential data respectively.

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